

GROTLI III, Norway

WMO: 013610

Lat: 62.016N

Lon: 7.664E

Elev: 2861

StdP: 13.24

Time Zone: 1.00 (EUC)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.9	-15.6	-25.3	1.5	-19.8	-20.8	1.9	-15.5	25.4	21.7	23.1	20.9	1.0	340	0.769

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.4	69.6	54.6	65.9	53.3	62.3	51.9	56.5	65.8	54.9	63.5	53.2	60.8	7.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
52.3	64.7	59.6	50.4	60.4	58.1	48.8	56.8	57.0	25.3	65.7	24.2	63.9	23.1	60.9	63.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
19.8	17.1	15.1	DB	-25.1	73.9	7.2	4.1	-30.2	76.8	-34.5	79.2	-38.5	81.5	-43.8	84.4
			WB	-25.4	58.3	7.1	2.5	-30.5	60.1	-34.7	61.6	-38.7	63.0	-43.9	64.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	32.2	14.9	16.9	22.1	29.4	38.3	45.4	51.6	49.3	43.4	32.8	23.4	17.6	(6)
	DBStd	15.70	13.22	12.19	10.23	8.62	6.90	5.21	5.61	4.96	5.82	7.44	10.75	13.37	(7)
	HDD50	6690	1089	928	866	618	368	155	48	74	207	533	798	1006	(8)
	HDD65	11983	1554	1348	1331	1068	829	589	416	485	647	997	1248	1471	(9)
	CDD50	185	0	0	0	0	4	17	98	54	11	1	0	0	(10)
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
	CDH74	20	0	0	0	0	0	0	20	0	0	0	0	0	(12)
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
	WSAvg	6.9	8.0	7.4	8.4	6.5	6.5	7.0	6.1	6.0	6.7	7.1	6.8	7.0	(14)
(15) Precipitation	PrecAvg	38.9	4.1	3.7	3.9	1.9	1.8	2.6	2.6	2.8	3.5	3.8	3.8	4.4	(15)
	PrecMax	53.1	10.9	7.3	13.5	7.1	4.1	5.9	4.3	6.4	10.5	8.5	10.3	8.3	(16)
	PrecMin	24.7	0.0	0.0	0.6	0.2	0.4	0.5	0.9	0.4	0.9	0.7	0.1	0.6	(17)
	PrecStd	7.2	3.1	2.2	3.0	1.4	0.8	1.3	0.9	1.3	2.3	1.9	3.1	2.1	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.7	41.3	42.2	50.7	67.7	69.6	77.4	68.5	63.9	56.3	45.1	42.0	(19)
		MCWB	35.0	35.8	36.7	41.1	52.5	52.8	59.2	53.8	50.6	48.9	40.3	37.9	(20)
	2%	DB	36.7	37.7	39.7	46.6	60.2	65.2	72.2	65.4	60.0	49.9	41.8	38.3	(21)
		MCWB	33.8	33.6	35.5	39.2	47.8	51.2	56.1	52.8	50.4	44.0	38.0	35.2	(22)
	5%	DB	34.3	34.9	37.6	43.5	54.8	60.2	68.3	62.2	56.7	46.3	38.9	35.8	(23)
		MCWB	31.6	31.4	33.8	37.0	45.4	48.4	55.2	52.4	48.9	41.7	35.7	33.0	(24)
	10%	DB	31.4	32.2	35.1	41.0	51.2	57.1	64.3	59.2	53.6	42.9	35.9	33.0	(25)
		MCWB	29.0	29.4	31.7	35.4	43.1	48.0	53.6	51.1	47.5	38.3	33.3	30.6	(26)
	0.4%	WB	36.0	36.8	37.9	42.1	52.7	55.2	60.1	56.7	54.3	51.4	40.9	38.3	(27)
		MCDB	38.5	40.6	40.9	49.0	65.6	64.4	72.9	63.3	60.2	56.8	44.7	41.4	(28)
(29) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	34.1	33.9	35.9	39.5	48.6	52.9	57.8	55.1	51.3	44.5	38.5	35.5	(29)
		MCDB	36.5	37.0	39.3	45.3	57.8	62.1	68.7	62.4	57.0	48.8	41.6	38.2	(30)
	5%	WB	31.7	31.7	34.1	37.7	45.8	50.6	56.2	53.5	49.6	41.7	35.9	33.0	(31)
		MCDB	34.1	34.5	37.4	43.2	54.6	58.6	66.0	60.3	55.7	46.2	38.6	35.4	(32)
	10%	WB	29.1	29.4	32.0	35.7	43.3	48.2	54.5	51.9	47.7	38.9	33.4	30.5	(33)
		MCDB	31.3	32.1	34.7	40.2	50.6	55.7	63.3	58.1	53.6	42.6	35.7	32.7	(34)
	5% DB	MDBR	13.7	14.8	15.7	16.5	15.9	16.9	17.4	16.0	14.7	12.6	12.4	14.0	(35)
		MCDBR	12.7	11.8	13.0	16.9	23.5	26.2	27.1	21.4	21.0	16.7	12.1	11.7	(36)
		MCWBR	10.7	9.3	9.7	10.4	13.7	14.4	13.6	11.6	12.8	11.7	9.5	10.1	(37)
		MCWBR	12.6	11.7	12.7	15.1	23.4	22.2	23.6	18.5	18.6	14.0	12.1	12.0	(38)
(40) Clear-Sky Solar Irradiance	taub		0.204	0.254	0.288	0.319	0.324	0.323	0.334	0.324	0.295	0.272	0.210	0.177	(40)
		taud	2.035	2.184	2.158	2.138	2.203	2.316	2.383	2.411	2.473	2.348	2.062	1.918	(41)
	Ebn at Noon		159	228	260	274	283	286	279	272	262	221	152	104	(42)
		Edh at Noon	11	20	30	38	39	36	33	29	23	17	10	7	(43)
(44) All-Sky Solar Radiation	RadAvg		60	222	598	1133	1492	1544	1455	1115	675	309	89	36	(44)
	RadStd		12	53	103	147	150	153	138	109	89	46	18	4	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (41 neighbors)		N/A	+1.37	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page